Toward Spiking Equilibrium Propagation: A Preliminary Study of Energy-Based Learning with Spiking Neural Dynamics

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Abstract

Equilibrium Propagation (EP) provides a biologically inspired alternative to backpropagation for training energy-based neural networks [1]. However, most EP implementations rely on continuous-valued neuron dynamics, limiting their compatibility with spiking neural networks (SNNs).

In this work, we extend equilibrium propagation (EP) to spiking computation by replacing continuous hidden-state dynamics with leaky integrate-and-fire (LIF) neurons. Our implementation builds upon an open-source EP framework [2], enabling a temporal, event-driven formulation of energy-based learning in spiking neural systems.

Experiments show stable training behavior and provide insights into spike activity, equilibrium convergence, and sparsity. We further discuss current limitations and outline future directions for systematic evaluation.

CCS Concepts

• **Computing methodologies** → **Neural networks**.

Keywords

Spiking Neural Networks, Equilibrium Propagation, Energy-Based Models, Deep Hopfield Networks

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1 Introduction

Equilibrium Propagation (EP) was proposed as a biologically plausible alternative to backpropagation for energy-based neural networks [1]. Meanwhile, Spiking Neural Networks (SNNs) have attracted growing attention due to their event-driven computation and potential energy efficiency. Both EP and SNNs can be interpreted as dynamical systems evolving toward stable states, making their integration a natural direction for energy-based spiking computation.

Existing EP implementations mainly rely on continuous neuron dynamics. In this work, we investigate whether EP can be extended to spiking neurons by replacing hidden-layer dynamics with leaky integrate-and-fire (LIF) neurons.

SNN-based EP is appealing as it replaces continuous-rate dynamics with sparse spike events, potentially enabling more efficient computation and providing an event-driven interpretation of equilibrium dynamics.

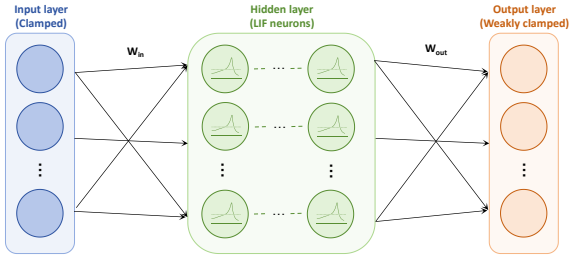


Figure 1: Overview of the proposed Equilibrium Propagation with Spiking Neural Networks (EP-SNN) framework.

2 Method

Figure 1 provides an overview of the proposed EP-SNN framework. The model extends standard equilibrium propagation by replacing continuous hidden-state dynamics with leaky integrate-and-fire (LIF) spiking neurons.

2.1 Baseline EP Framework

Our implementation builds upon an open-source energy-based learning framework developed by Scellier and collaborators[2]. In this framework, the network is defined as a recurrent system that evolves toward fixed points determined by an energy function.

2.2 Implementation of the Spiking EP Model

We replace the original energy function defined on hard-sigmoid hidden states with a spiking counterpart, where hidden-layer dynamics are governed by membrane potential integration and discrete spike events. In contrast to the original fixed-point minimization used in Deep Hopfield Network (DHN), we adopt a temporal LIF-based minimizer that runs for a fixed number of simulation steps per EP phase, and the output is computed from averaged firing rates.

Importantly, the EP learning rule, stochastic gradient descent optimizer, dataset pipeline, and monitoring components are kept

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identical to the original implementation, ensuring that any observed differences arise solely from the introduction of spiking neural dynamics.

The spiking model introduces additional hyperparameters, including membrane decay, firing threshold, and simulation length, which control the temporal behavior of the LIF dynamics while preserving the underlying equilibrium propagation framework.

2.3 Spiking Neural Dynamics

We replace continuous-valued hidden states with LIF spiking neurons, which model neurons as dynamical systems that integrate incoming currents over time and emit discrete spikes once a threshold is reached.

Each neuron evolves according to:

$$V_{t+1} = \lambda V_t + I_t - S_t V_{reset} \tag{1}$$

$$S_t = H(V_t - V_{th}) \tag{2}$$

where $H(\cdot)$ is the Heaviside step function. V_t denotes the membrane potential at time step t, I_t represents the synaptic input current, S_t is the binary spike output, V_{th} is the firing threshold, and V_{reset} denotes the reset value applied after a spike event. The parameter $\lambda \in (0, 1)$ is the membrane leakage factor controlling temporal decay.

2.4 Surrogate Gradient Learning

Since spike generation is non-differentiable, we adopt surrogate gradients to enable learning through spike-based dynamics.

To support learning, gradients are approximated by replacing the derivative of the spike function with a smooth surrogate:

$$\frac{\partial S_t}{\partial V_t} \approx \sigma'(V_t) \tag{3}$$

We use a piecewise linear (hard-sigmoid) surrogate defined as:

$$\sigma(V_t) = \text{clip}\left(\frac{V_t}{2V_{th}} + \frac{1}{2},\ 0,\ 1\right) \tag{4}$$

3 Preliminary Experiments

Experiments were conducted on the MNIST dataset.

To isolate the effect of spiking dynamics, all hyperparameters were kept identical to those of the original EP implementation, including learning rate, batch size, and training schedule.

Figure 2 compares the classification error during training for the original EP model and the proposed spiking version.

In addition to classification performance, we also monitored the mean spike activity of the hidden layer over a short evaluation window during training. The observed firing rate remained consistently low, typically in the range of 0.03–0.07, indicating sparse event-driven dynamics in the spiking network.

Preliminary observations include:

- Stable learning can be achieved with spike-based dynamics.
- The proposed model exhibits convergence behavior similar to the original EP implementation.
- Hidden-layer spike activity remains sparse throughout training.

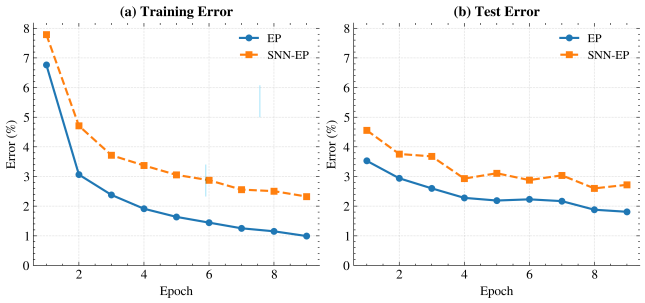


Figure 2: Training and test error comparison between EP and SNN-EP on MNIST.

- EP learning remains effective after introducing discrete spike events.

4 Discussion and Future Work

Extending equilibrium propagation (EP) to spiking neural systems raises several open questions:

- How should equilibrium be defined in a temporal, event-driven spiking system?
- How does spike sparsity influence EP learning stability and convergence?
- What is the relationship between firing rate, energy efficiency, and classification performance?

A key motivation of this work is the potential energy efficiency of spiking implementations due to sparse, event-driven computation. In this study, efficiency is only indirectly reflected through low firing rates, rather than explicit hardware-level energy measurement.

Future work will extend the framework to larger datasets, deeper architectures, and temporal input settings, and further explore benchmarking on neuromorphic hardware platforms.

5 Conclusion

This work presents a preliminary exploration of integrating spiking neural dynamics into Equilibrium Propagation. By replacing continuous hidden states with LIF neurons and employing surrogate gradients, we demonstrate the feasibility of training energy-based spiking networks within an EP framework. The results provide an initial step toward bridging equilibrium-based learning and spiking computation.

References

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